ATX power supply

5V (regulated)

GND

12V

GND

Breadboard

BB 5V rail

Potentiometer
wiper 0–3.3 V

BB GND rail

W03 analog in, 0–3.3 V

UNO power board

Power

UNO32 IO board

W03: analog in (AD_PORTW3)

Y06: servo out (RC_PORTY06, Timer4)

LED bank: scaled pot value (LED_BANK1–3)

Y06 digital out, 0–3.3 V

pulse 1–2 ms every 50 ms

RC Servo

5V (orange)

GND (brown)

Signal (yellow)

5 V regulated, max 5 V

Common ground: ATX, UNO boards, breadboard rails, and servo

UCSC ECE118 – Student: Amber

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Title: ECE118 Lab 3 – Part 1: Driving an RC Servo (Block Diagram)

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